



ATLIQ HARDWARE CONSUMER GOODS

A DATA-DRIVEN ANALYSIS ADDRESSING IO BUISNESS AD
HOC REQUEST USING SQL.

Tool Used: SQL, PowerPoint



Agenda



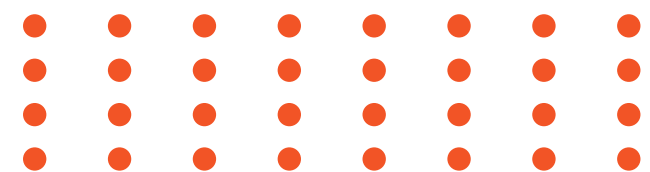
01 Introduction

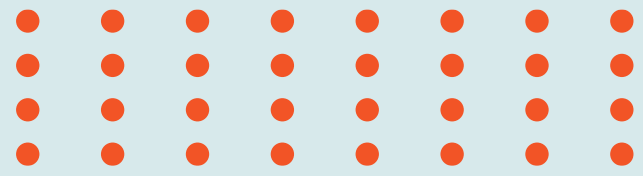
02 Dataset Overview

03 Ad-Hoc Questions

04 SQL Analysis

05 Insights & Recommendations





Introduction



AtliQ Hardware is a leading consumer goods company seeking data-driven insights to improve business decision-making. In this project, I analyzed company sales, product, and customer data to answer 10 ad-hoc business requests using SQL. The objective was to extract meaningful insights, identify trends, and support strategic planning through structured querying and data analysis.

Dataset Overview



Database Purpose: Comprehensive hardware sales and product database containing customer information, product details, pricing, manufacturing costs, and monthly sales transactions for fiscal years 2020-2021.

CORE TABLES (6)

dim_customer

Customer master data including codes, names, platform, channel, market, and geographic region information.

fact_gross_price

Initial product pricing before deductions or taxes for fiscal years 2020-2021.

fact_pre_invoice_deduction

Pre-invoice discount percentages applied before invoice generation for large orders.

dim_product

Product master data with codes, divisions, segments, categories, and variants (Standard, Plus, Premium).

fact_manufacturing_cost

Production costs including raw materials, labor, and overhead expenses by product and fiscal year.

fact_sales_monthly

Monthly sales transactions including date, product, customer, quantity sold, and fiscal year.



AD-HOC BUSINESS QUESTIONS - ATLIQ HARDWARE

Q1. Provide the list of markets in which customer "AtliQ Exclusive" operates its business in the APAC Region.

Q2. What is the percentage of unique product increase in 2021 vs. 2020? The final output contains these fields Unique Products 2020, Unique Products 2021 and Percentage Change.

Q3. Provide a report with all the unique product counts for each segment and sort them in descending order of product counts. The final output contains 2 fields, segment, product count.

Q4. Which segment had the most increase in unique products in 2021 vs 2020? The final output contains these fields, segment, product_count_2020, product_count_2021, difference.

Q5. Get the products that have the highest and lowest manufacturing costs. The final output should contain these fields - Product Code, Product and Manufacturing Cost.

Q6. Generate a report which contains the top 5 customers who received an average high pre invoice discount pct for the fiscal year 2021 and in the Indian market. The final output contains these fields – Customer Code, Customer, Avg Discount Pct

Q7. Get the complete report of the Gross sales amount for the customer "AtliQ Exclusive" for each month. This analysis helps to get an idea of low and high-performing months and take strategic decisions. The final report contains these columns: Month, Year, Gross sales.

Q8. In which quarter of 2020, got the maximum total sold quantity? The final output contains these fields sorted by the total sold quantity, Quarter, total sold quantity

Q9. Which channel helped to bring more gross sales in the fiscal year 2021 and the percentage of contribution? The final output contains these fields, channel, gross sales mln, percentage

Q10. Get the Top 3 Products in each division that have a high total sold quantity in the fiscal year? The final output contains these fields – Division, Product Code, Product, Total Sold Quantity, Rank Order.

Q1. Provide the list of markets in which customer "AtliQ Exclusive" operates its business in the APAC Region.



Query

```
SELECT DISTINCT
    customer,
    market,
    region
FROM dim_customer
WHERE region = 'APAC'
AND customer = 'AtliQ Exclusive';
```

Output

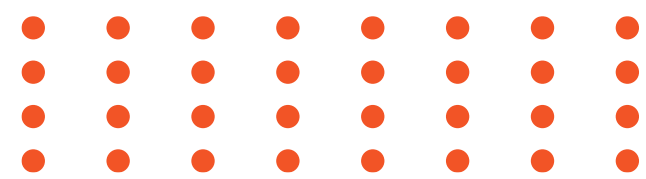
customer	market	region
AtliQ Exclusive	India	APAC
AtliQ Exclusive	Indonesia	APAC
AtliQ Exclusive	Japan	APAC
AtliQ Exclusive	Philippines	APAC
AtliQ Exclusive	South Korea	APAC
AtliQ Exclusive	Australia	APAC
AtliQ Exclusive	Newzealand	APAC
AtliQ Exclusive	Bangladesh	APAC

Insights:

ATLIQ EXCLUSIVE HAS A STRONG PRESENCE ACROSS THE APAC REGION, OPERATING IN 8 KEY MARKETS INCLUDING INDIA, INDONESIA, JAPAN, SOUTH KOREA, AUSTRALIA, PHILLIPHINES AND EMERGING MARKETS LIKE BANGLADESH AND THE NEWZELAND.



Q2. What is the percentage of unique product increase in 2021 vs. 2020? The final output contains these fields Unique Products 2020, Unique Products 2021 and Percentage Change.



Query

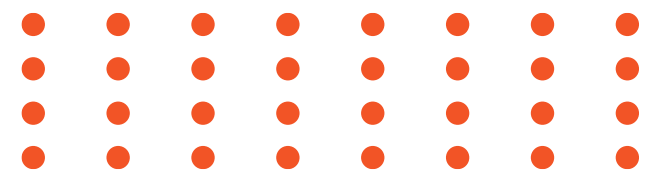
```
WITH CTE AS(  
SELECT  
COUNT(DISTINCT CASE WHEN fiscal_year = 2020 THEN product_code END) AS unique_product_2020,  
COUNT(DISTINCT CASE WHEN fiscal_year = 2021 THEN product_code END) AS unique_product_2021  
FROM fact_sales_monthly)  
  
SELECT  
unique_product_2020,  
unique_product_2021,  
CONCAT(ROUND(((unique_product_2021 - unique_product_2020) * 100 / unique_product_2020),2),"%") AS percentage_change  
FROM CTE
```

Output

unique_product_2020	unique_product_2021	percentage_change
245	334	36.33%

Insights:

THE NUMBER OF UNIQUE PRODUCTS INCREASED FROM 245 IN 2020 TO 334 IN 2021, REPRESENTING A 36.33% GROWTH, WHICH INDICATES A SIGNIFICANT EXPANSION IN THE PRODUCT PORTFOLIO TO SUPPORT BUSINESS GROWTH AND MARKET DEMAND.



Q3. Provide a report with all the unique product counts for each segment and sort them in descending order of product counts. The final output contains 2 fields, segment, product count.



Query

```
SELECT
    segment, COUNT(*) AS product_count
FROM
    dim_product
GROUP BY segment
ORDER BY product_count DESC;
```

Output

segment	product_count
Notebook	129
Accessories	116
Peripherals	84
Desktop	32
Storage	27
Networking	9

Insights:

THE NOTEBOOK SEGMENT LEADS WITH THE HIGHEST NUMBER OF UNIQUE PRODUCTS (129), FOLLOWED BY ACCESSORIES (116), INDICATING THESE SEGMENTS ARE THE PRIMARY FOCUS OF ATLIQ’S PRODUCT PORTFOLIO, WHILE NETWORKING HAS THE SMALLEST PRODUCT RANGE.



Q4. Which segment had the most increase in unique products in 2021 vs 2020? The final output contains these fields, segment, product_count_2020, product_count_2021, difference.



Query

```
WITH CTE AS(
  SELECT p.segment as segment,
  COUNT(DISTINCT CASE WHEN gp.fiscal_year = 2020 THEN p.product_code END) AS unique_product_2020,
  COUNT(DISTINCT CASE WHEN gp.fiscal_year = 2021 THEN p.product_code END) AS unique_product_2021
  FROM dim_product p
  JOIN fact_gross_price gp
  ON p.product_code = gp.product_code
  GROUP BY p.segment)

SELECT segment,
  unique_product_2020,
  unique_product_2021,
  (unique_product_2021 - unique_product_2020) AS difference
FROM CTE
ORDER BY difference DESC;
```

Output

segment	unique_product_2020	unique_product_2021	difference
Accessories	69	103	34
Notebook	92	108	16
Peripherals	59	75	16
Desktop	7	22	15
Storage	12	17	5
Networking	6	9	3

Insights:

THE ACCESSORIES SEGMENT RECORDED THE HIGHEST INCREASE IN UNIQUE PRODUCTS, GROWING FROM 69 IN 2020 TO 103 IN 2021 (+34), INDICATING THE STRONGEST PRODUCT EXPANSION AMONG ALL SEGMENTS.



Q5. Get the products that have the highest and lowest manufacturing costs. The final output should contain these fields - Product Code, Product and Manufacturing Cost.



Query

```
SELECT
    p.product_code AS Product_Code,
    p.product AS Product,
    c.manufacturing_cost AS Manufacturing_Cost
FROM
    dim_product p
    JOIN
    fact_manufacturing_cost c ON p.product_code = c.product_code
WHERE
    c.manufacturing_cost IN (SELECT
        MAX(manufacturing_cost)
        FROM
            fact_manufacturing_cost UNION SELECT
        MIN(manufacturing_cost)
        FROM
            fact_manufacturing_cost)
ORDER BY manufacturing_cost DESC;
```

Output

Product_Code	Product	Manufacturing_Cost
A6120110206	AQ HOME Allin1 Gen 2	240.5364
A2118150101	AQ Master wired x1 Ms	0.8920

Insights:

THERE IS A SIGNIFICANT VARIATION IN MANUFACTURING COST ACROSS PRODUCTS. AQ HOME ALLIN1 GEN 2 HAS THE HIGHEST MANUFACTURING COST (240.54), WHILE AQ MASTER WIRED X1 MS HAS THE LOWEST COST (0.89), HIGHLIGHTING OPPORTUNITIES FOR COST OPTIMIZATION AND PRICING STRATEGY REVIEW ACROSS THE PRODUCT PORTFOLIO.



Q6. Generate a report which contains the top 5 customers who received an average high pre invoice discount pct for the fiscal year 2021 in the Indian market. The final output contains these fields – Customer Code, Customer, Avg Discount Pct



Query

```
SELECT
  c.customer_code AS Customer_Code,
  c.customer AS customer,
  ROUND(AVG(p.pre_invoice_discount_pct) * 100, 2) AS AVG_Discount_Pct
FROM
  dim_customer c
  JOIN
  fact_pre_invoice_deductions p ON c.customer_code = p.customer_code
WHERE
  p.fiscal_year = 2021
  AND c.market = 'India'
GROUP BY c.customer_code , c.customer
ORDER BY AVG_Discount_Pct DESC
LIMIT 5;
```

Output

Customer_Code	customer	AVG_Discount_Pct
90002009	Flipkart	30.83
90002006	Viveks	30.38
90002003	Ezone	30.28
90002002	Croma	30.25
90002016	Amazon	29.33

Insights:

IN FY 2021, MAJOR INDIAN RETAILERS SUCH AS FLIPKART, VIVEKS, EZONE, CROMA, AND AMAZON RECEIVED THE HIGHEST AVERAGE PRE-INVOICE DISCOUNTS (~29-31%), INDICATING A HEAVY DISCOUNTING STRATEGY FOR KEY HIGH-VOLUME CUSTOMERS TO DRIVE SALES IN THE INDIAN MARKET.



Q7. Get the complete report of the Gross sales amount for the customer “AtliQ Exclusive” for each month . This analysis helps to get an idea of low and high-performing months and take strategic decisions. The final report contains these columns: Month, Year, Gross sales.



Query

```
SELECT
  DATE_FORMAT(FS.date, '%Y %b') AS YearMonth,
  CONCAT(ROUND(SUM(G.gross_price * FS.sold_quantity) / 1000000,
              2),
         'M') AS Gross_Sales_Amount
FROM
  fact_sales_monthly FS
  JOIN
  dim_customer C ON FS.customer_code = C.customer_code
  JOIN
  fact_gross_price G ON FS.product_code = G.product_code
WHERE
  C.customer = 'AtliQ Exclusive'
GROUP BY 1
ORDER BY Gross_Sales_Amount DESC;
```

Output

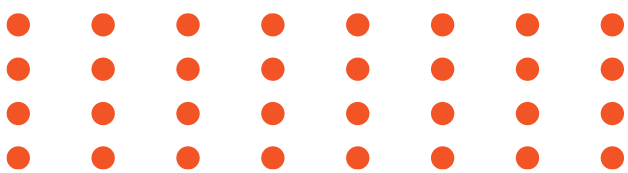
	YearMonth	Gross_Sales_Amount
▶	2019 Dec	9.76M
	2020 Jan	9.58M
	2019 Sep	9.09M
	2020 Feb	8.08M
	2020 Aug	5.64M
	2020 Jul	5.15M
	2020 Nov	32.25M
	2020 Jun	3.43M
	2020 Oct	21.02M
	2020 Dec	20.41M
	2021 Jan	19.57M
	2020 Sep	19.53M
	2021 May	19.20M
	2021 Mar	19.15M
	2021 Jul	19.04M
	2021 Feb	15.99M
	2021 Jun	15.46M
	2019 Nov	15.23M
	2021 Apr	11.48M
	2021 Aug	11.32M
	2019 Oct	10.38M
	2020 May	1.59M
	2020 Apr	0.80M
	2020 Mar	0.77M

Insights:

AtliQ Exclusive’s gross sales show a strong upward trend from 2020 to 2021, with peak performance in November 2020 (32.25M) and consistently high sales across most months in 2021 (~15-20M), while April-May 2020 marked the lowest sales period, indicating a temporary demand slowdown.



Q8. In which quarter of 2020, got the maximum total sold quantity? The final output contains these fields sorted by the total sold quantity, Quarter, total sold quantity



Query

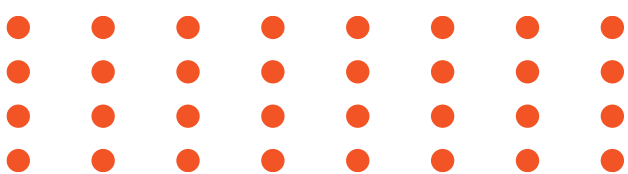
```
SELECT
    QUARTER(date) AS Quarter,
    CONCAT(SUM(sold_quantity) / 1000000, 'M') AS Sold_Quantity
FROM
    fact_sales_monthly
GROUP BY QUARTER(date)
ORDER BY Sold_Quantity;
```

Output

Quarter	Sold_Quantity
2	14.2272M
1	14.5658M
3	16.2716M
4	25.8729M

Insights:

In 2020, Q4 recorded the highest total sold quantity (25.87M), significantly outperforming other quarters, indicating strong year-end demand, while Q2 and Q1 showed comparatively lower sales volumes, reflecting weaker mid-year performance.



Q9. Which channel helped to bring more gross sales in the fiscal year 2021 and the percentage of contribution? The final output contains these fields, channel, gross sales mln, percentage



Query

```
WITH CTE AS (  
  SELECT  
    c.channel AS Channel,  
    SUM(p.gross_price * s.sold_quantity) / 1000000 AS Gross_Sales_Mln  
  FROM dim_customer c  
  JOIN fact_sales_monthly s  
    ON c.customer_code = s.customer_code  
  JOIN fact_gross_price p  
    ON p.product_code = s.product_code  
  WHERE s.fiscal_year = 2021  
  GROUP BY c.channel  
)  
  
SELECT  
  Channel,  
  CONCAT(ROUND(Gross_Sales_Mln, 2), 'M') AS gross_sales_mln,  
  ROUND(  
    (Gross_Sales_Mln / (SELECT SUM(Gross_Sales_Mln) FROM CTE)) * 100,  
    2  
  ) AS percentage  
FROM CTE  
ORDER BY Gross_Sales_Mln DESC;
```

Output

Channel	gross_sales_mln	percentage
Direct	406.69M	15.47
Distributor	297.18M	11.31
Retailer	1924.17M	73.22

Insights:

In FY 2021, the Retailer channel was the primary revenue driver, contributing 73.22% (1924.17M) of total gross sales, significantly outperforming Direct (15.47%) and Distributor (11.31%) channels, highlighting a strong dependency on retail partnerships for revenue growth.



Q10. Get the Top 3 Products in each division that have a high total sold quantity in the fiscal year? The final output contains these fields – Division, Product Code, Product, Total Sold Quantity, Rank Order.



Query

```
WITH CTE AS (  
  SELECT  
    p.division AS Division,  
    p.product_code AS Product_Code,  
    p.product AS Product,  
    SUM(s.sold_quantity) AS Total_Sold_Quantity,  
    DENSE_RANK() OVER(PARTITION BY Division ORDER BY SUM(s.sold_quantity) DESC) AS Rank_Order  
  FROM dim_product p  
  JOIN fact_sales_monthly s  
  ON p.product_code = s.product_code  
  WHERE s.fiscal_year = 2021  
  GROUP BY p.division, p.product_code, p.product)  
  
SELECT *  
FROM CTE  
WHERE Rank_Order <= 3;
```

Output

Division	Product_Code	Product	Total_Sold_Quantity	Rank_Order
N & S	A6720160103	AQ Pen Drive 2 IN 1	701373	1
N & S	A6818160202	AQ Pen Drive DRC	688003	2
N & S	A6819160203	AQ Pen Drive DRC	676245	3
P & A	A2319150302	AQ Gamers Ms	428498	1
P & A	A2520150501	AQ Maxima Ms	419865	2
P & A	A2520150504	AQ Maxima Ms	419471	3
PC	A4218110202	AQ Digit	17434	1
PC	A4319110306	AQ Velocity	17280	2
PC	A4218110208	AQ Digit	17275	3

Insights:

In FY 2021, each division’s sales were driven by a small set of high-performing products–N & S was led by AQ Pen Drive 2 IN 1, P & A by AQ Gamers Ms, and PC by AQ Digit / AQ Velocity–indicating that top 3 products in every division contribute a major share of total sold quantity, highlighting opportunities to prioritize inventory, promotions, and supply planning around these key SKUs.



Strategic Recommendations

Data-Driven Insights for Business Growth & Optimization



1

OPTIMIZE RETAILER CHANNEL STRATEGY

Retailers drive 73.22% of FY 2021 revenue. Strengthen retail partnerships through exclusive agreements, co-marketing initiatives, and dedicated account management.

→ Priority: High | Expected Impact: +5-8% Revenue

2

EXPAND HIGH-GROWTH SEGMENTS

Accessories segment grew fastest (+34 products in 2021). Allocate more R&D budget and inventory to capitalize on this 36.33% overall product portfolio growth.

→ Priority: High | Expected Impact: +12-15% Growth

3

SKU RATIONALIZATION & FOCUS

Top 3 products per division drive majority of sales. Consolidate marketing spend and production on high-performing SKUs while evaluating underperforming products for discontinuation.

→ Priority: Medium | Expected Impact: +8-10% Margin

4

REVISE DISCOUNT STRATEGY

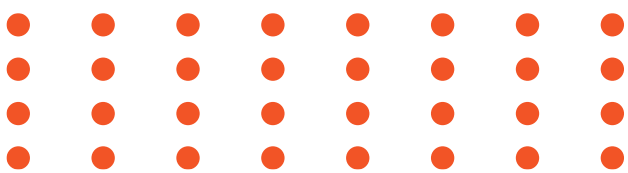
Indian retailers receive 29-31% pre-invoice discounts. Analyze discount elasticity—implement tiered discounting based on volume/loyalty to improve margins while maintaining competitive positioning.

→ Priority: High | Expected Impact: +3-5% Margin



Strategic Recommendations

Data-Driven Insights for Business Growth & Optimization



1 OPTIMIZE RETAILER CHANNEL STRATEGY

Retailers drive 73.22% of FY 2021 revenue. Strengthen retail partnerships through exclusive agreements, co-marketing initiatives, and dedicated account management.

→ Priority: High | Expected Impact: +5-8% Revenue

2 EXPAND HIGH-GROWTH SEGMENTS

Accessories segment grew fastest (+34 products in 2021). Allocate more R&D budget and inventory to capitalize on this 36.33% overall product portfolio growth.

→ Priority: High | Expected Impact: +12-15% Growth

3 SKU RATIONALIZATION & FOCUS

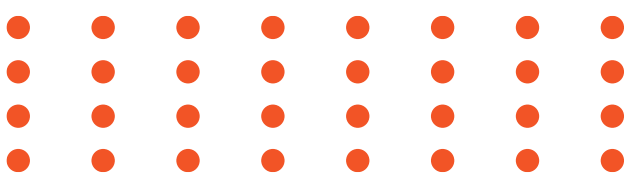
Top 3 products per division drive majority of sales. Consolidate marketing spend and production on high-performing SKUs while evaluating underperforming products for discontinuation.

→ Priority: Medium | Expected Impact: +8-10% Margin

4 REVISE DISCOUNT STRATEGY

Indian retailers receive 29-31% pre-invoice discounts. Analyze discount elasticity—implement tiered discounting based on volume/loyalty to improve margins while maintaining competitive positioning.

→ Priority: High | Expected Impact: +3-5% Margin



THANK
YOU

